EX.NO: 3	Survey report on Emerging Trends in low-code application Development

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Abstract:

Recent research shows that low-code development is moving beyond simple drag-and-drop application building toward **AI-assisted development, citizen development, automation, cloud integration, governance, and enterprise-scale applications**. Gartner's 2025 forecast, for example, projects the low-code development technologies market to reach **$58.2 billion by 2029**, with AI agents and citizen development among the major adoption drivers.

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1. Introduction

Low-code application development is a modern software development approach that uses visual interfaces, drag-and-drop tools, reusable components, and minimal coding. It enables both developers and non-technical users to build applications faster. Modern low-code platforms also integrate AI, automation, data, and cloud services, making application development easier and more efficient.

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2.Overview of Low-Code Application Development

The evolution of application development can be represented as:

Traditional Coding → RAD → Low-Code → No-Code → AI-Assisted Low-Code → AI Agent-Based Development

Traditional Development

Developers manually write code using languages such as:

- Java
- Python
- C#
- JavaScript
- PHP

Low-Code Development

Developers use:

- Drag-and-drop interfaces
- Visual workflows
- Pre-built components
- Database connectors
- APIs
- Reusable templates

AI-Assisted Low-Code

The newest stage adds:

- Natural-language prompts
- AI-generated application components
- AI-assisted workflows
- Automated data processing
- AI agents
- Intelligent recommendations

Gartner's 2026 market analysis notes that leading low-code platforms are increasingly embedding **agentic AI directly into development environments**.

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3. Objectives and Research Methodology

Objectives

The main objectives of this survey are:

- To understand awareness of low-code development.
- To identify emerging trends in low-code technology.
- To study the benefits and challenges of low-code platforms.
- To understand the role of AI in low-code development.
- To analyze users' future interest in low-code platforms.

Research Methodology

The study uses a **survey-based research method**. Data can be collected using **Google Forms** from students, developers, faculty, and IT professionals. The collected responses are analyzed using **frequency, percentage, tables, charts, and graphs**.

4. Emerging Trends in Low-Code Development

Low-code development is evolving rapidly with new technologies.

Major emerging trends:

AI-Assisted Development: AI helps generate applications, workflows, formulas, and code from natural-language instructions.

Citizen Development: Non-programmers can create applications for their business requirements.

Workflow Automation: Repetitive tasks such as approvals, notifications, and data entry can be automated.

Cloud Integration: Applications can easily connect with cloud databases and services.

Data Analytics: Low-code platforms provide dashboards, reports, and data visualization.

AI Agents: Newer platforms are moving toward AI agents that can perform multi-step tasks with less manual configuration.

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5. Survey Questionnaire and Data Collection

A questionnaire is prepared to understand people's awareness, experience, opinions, and expectations regarding low-code development.

Sample questions:

1. Are you aware of low-code development?
2. Have you used a low-code platform?
3. Which low-code platform have you used?
4. What is the main benefit of low-code development?
5. Which emerging trend is most important?
6. What is the biggest challenge?
7. Do you think AI will increase low-code adoption?
8. Would you use low-code platforms in future?

The survey responses are collected through **Google Forms** and organized for further analysis.

6. Data Analysis and Statistical Interpretation

The collected survey data is analyzed to identify patterns and trends.

Percentage

Percentage is calculated using:

Percentage = (Number of responses / Total responses) × 100

Frequency

Frequency represents the **number of respondents** selecting a particular answer.

Mean

Mean can be used for questions based on a rating scale such as:

1 = Strongly Disagree → 5 = Strongly Agree

Example

If 70 out of 100 respondents are aware of low-code development:

Percentage = (70 / 100) × 100 = 70%

The results help identify the overall awareness, adoption, benefits, and concerns related to low-code development.

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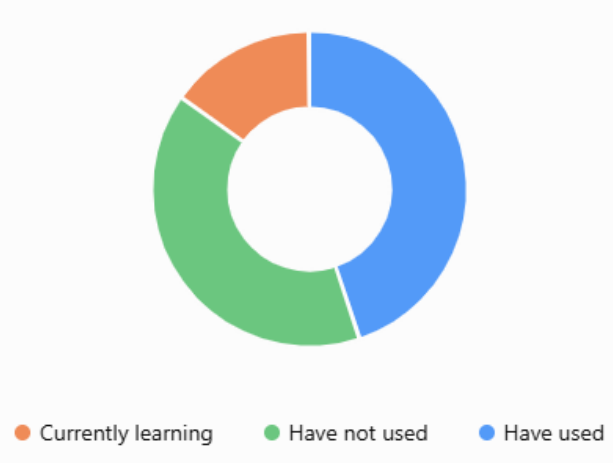
7. Charts and Graphical Representation

Charts make the survey results easier to understand and compare.

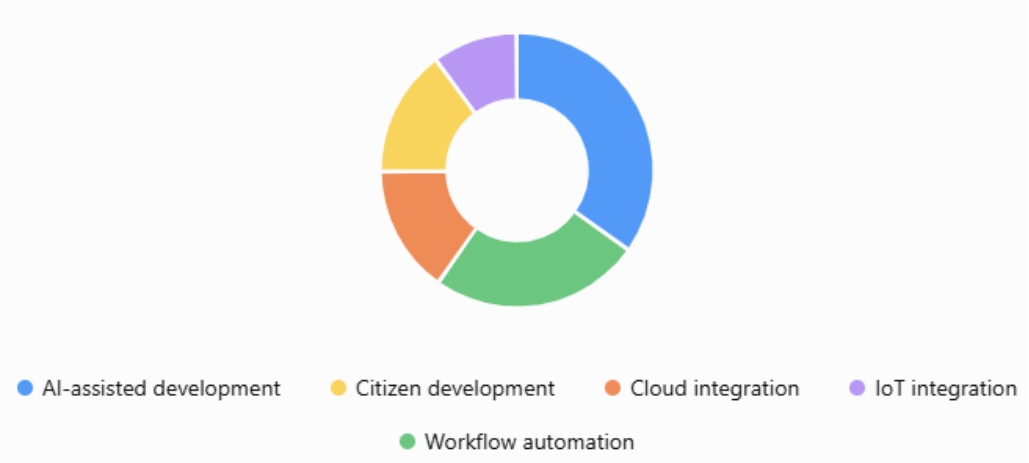
Awareness of Low-Code Development



Experience with Low-Code



Most Important Emerging Trend



8. Benefits and Challenges of Low-Code Development

Benefits

- Faster application development
- Reduced coding effort
- Lower development cost
- Easy prototyping
- Workflow automation
- Improved productivity
- Easier collaboration

- Greater participation from non-programmers

Challenges

- Security concerns
- Vendor lock-in
- Limited customization for complex applications
- Scalability issues
- Governance requirements
- Dependence on the platform
- Need for skilled developers for advanced applications

Therefore, low-code provides speed and simplicity but still requires proper planning and governance.

9. Real-World Applications and Future Trends

Low-code platforms are used in many areas such as:

- **Healthcare:** Patient management systems
- **Education:** Student management applications
- **Banking:** Internal workflow applications
- **Retail:** Inventory and sales management
- **Manufacturing:** Production monitoring
- **Human Resources:** Employee management
- **Business:** Customer and process management

Future Trends

The future of low-code development is expected to focus on:

AI + Low-Code + Cloud + Automation + Data Analytics + AI Agents

AI will increasingly help users create applications through natural-language instructions, while low-code platforms will continue to support faster business application development.

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10. Findings and Conclusion

Findings

The survey can be used to identify:

- The level of awareness of low-code development.
- The most commonly used platforms.
- The most important emerging trends.
- Major benefits expected by users.
- Main challenges affecting adoption.
- The expected impact of AI.
- Users' willingness to adopt low-code platforms in the future.

Comparison of Different Low-Code Platforms

Platform	Main Use	Easy to Use	Customization	Integration	Best Suitable For
Zoho Creator	Creating business applications	Very Easy	High	Good	Small businesses and students
Microsoft Power Apps	Business apps and automation	Very Easy	High	Excellent	Organizations using Microsoft products
OutSystems	Enterprise application development	Easy	Very High	Excellent	Large organizations
Mendix	Web and mobile applications	Easy	Very High	Excellent	Enterprise-level applications
Appian	Workflow and process automation	Easy	High	Excellent	Business process management
Salesforce Platform	CRM and customer applications	Easy	High	Excellent	Sales and customer management

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Conclusion

Low-code platforms have made software development easier, faster, and more accessible. Platforms such as Zoho Creator, Microsoft Power Apps, OutSystems, Mendix, Appian, and Salesforce provide visual tools, drag-and-drop features, workflow automation, and ready-made components that reduce the need for writing large amounts of code.

Each platform has its own strengths. Zoho Creator and Power Apps are easy to use and suitable for beginners and small businesses, while OutSystems and Mendix are better for large and complex applications. Appian is mainly useful for process automation, whereas Salesforce is widely used for CRM and customer-related applications.

Overall, **low-code platforms help organizations reduce development time, cost, and effort while allowing applications to be built quickly and efficiently**. However, traditional programming is still important for highly customized and complex systems. Therefore, the best choice of platform depends on the organization's requirements, budget, technical skills, and application complexity.

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